Arinze Ogbonna

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SUMMARY

Fullstack Software Engineer with 8 years of experience building high-performance web applications and data-intensive systems across healthcare, fintech, and civic tech. Advanced proficiency in TypeScript, Node.js, React, and Next.js, with hands-on experience in observability tooling, real-time data streaming, performance profiling, and large-scale data processing. Proven track record of building scalable web services, monitoring dashboards, and CI/CD pipelines, with a focus on application performance, developer experience, and production reliability.

CORE COMPETENCIES

- **Frontend & Frameworks:** React, Next.js, TypeScript, Vue.js, Redux, Zustand, React Query, and SSR/SSG/ISR.
- **Performance & Observability:** Lighthouse, Core Web Vitals, React Profiler, bundle analysis (webpack/rollup), Sentry, Datadog RUM, Langfuse, and Allure Reports.
- **Backend & Data:** NodeJS, NestJS, REST APIs, GraphQL, PostgreSQL (pgvector), MongoDB, Prisma, and JWT authentication.
- **Real-Time & Streaming:** SSE streaming, EventSource API, WebSockets, keepalive heartbeats, and real-time data processing.
- **Data Visualization:** Recharts (22 chart types), react-chartjs, dynamic dashboards, and interactive data UIs.
- **Cloud & DevOps:** AWS (CloudFront, S3), Vercel, Docker, GitHub Actions, CI/CD pipelines, and Nx monorepo orchestration.
- **Testing & Quality:** Playwright, Jest, Cypress, Storybook, ESLint, Prettier, and CI-driven QA.

WORK EXPERIENCE

Genentech (Roche) — San Francisco, CA (Remote)

Frontend Engineer — *May 2023 - Present*

- Spearheaded the redesign of the main clinical trial data product, migrating from a Vue.js web application to a React Chrome extension, optimizing workflows for clinical site integrations and enhancing user adoption.
- Optimized application performance by implementing lazy loading, code splitting, dynamic imports, and CDN asset delivery, reducing load times significantly as measured via Lighthouse and Chrome DevTools profiling.
- Co-developed a full-stack Next.js chat widget powered by a RAG system using OpenAI GPT models, LangChain, Vercel AI SDK, and Chroma vector database, enabling users to query their data in real time via a conversational interface.
- Introduced Redux for state management and React Query for request caching, refactoring legacy class components to functional components and streamlining data flow across the application.
- Increased release reliability by integrating Playwright end-to-end tests into the CI/CD pipeline, automating reporting with Allure Reports on GitHub Pages and configuring Slack notifications for build quality visibility.
- Implemented JWT-based authentication and role-based access control in Next.js with TypeScript and React Query, using HTTP-only cookies for secure token handling and permission-based UI rendering.
- Architected a reusable design system in React and Radix UI with Tailwind CSS, improving UI consistency, accessibility (WCAG), and developer velocity across multiple product teams.
- Mentored a team of two junior developers (interns), fostering a culture of technical excellence and code quality.

Descasio — Lagos, Nigeria (Hybrid)

Senior Frontend Developer — *May 2021 - Mar 2023*

- Optimized frontend performance by profiling render bottlenecks with React Profiler and Lighthouse, reducing bundle size by 30-40% through rollup-plugin-visualizer and webpack-bundle-analyzer, achieving a 35% faster load time.
- Led the design and implementation of an intuitive React user interface, achieving a 25% boost in user engagement and collaborating with architects to create a scalable frontend architecture that accelerated development by 30%.
- Drove a 40% improvement in application performance by optimizing caching mechanisms across the React application.
- Enhanced development workflow by introducing Jest and Enzyme testing frameworks, leading to a 20% decrease in bug reports.
- Established engineering standards using ESLint, Prettier, Husky, Lint-staged, and GitHub Actions, reducing errors and accelerating CI/CD processes.
- Mentored junior engineers on frontend best practices, fostering high-quality, maintainable code across projects.

ActivEdge Technologies — Lagos, Nigeria

Senior React Developer — *Oct 2019 - Sep 2021*

- Built a high-volume bank Settlement Solution in a team of 3, handling thousands to millions of daily transactions and boosting settlement rates by 80%.

- Adopted the native EventSource API for real-time data streaming, replacing socket.io for faster updates and a more responsive interface, processing high-frequency transaction data.
- Built dynamic data visualization dashboards using React and react-chartjs with Redux Toolkit and Ant Design, enabling real-time monitoring of transaction flows and settlement statuses.
- Developed critical payment modules (MasterCard, VISA, Express-cash, E-commerce) using React, Redux, RxJS, and Axios.
- Built a real-time account reconciliation solution, boosting daily reconciliation by 60% through innovative transaction processing strategies.

Founders Lair Beijing, China (Remote)
Full Stack Engineer (Contract) *Oct 2020 - Feb 2021*

- Developed NodeJS RESTful APIs and translated business requirements into efficient MongoDB database models and schemas.
- Transformed Figma designs into reusable React components and engineered a CI/CD pipeline for automated deployment to Heroku and AWS.
- Mentored two interns through frontend and backend projects, providing technical guidance and task management.

SureTrade Business Solutions Nigeria
Frontend Developer *Mar 2018 - Oct 2019*

- Designed MongoDB database schemas and implemented server-side logic in NodeJS, building and testing REST API endpoints for multiple client-facing applications.
- Developed SureEdu, a school management portal in a team of 5, handling staff payroll and attendance tracking, student records (exams, reports, attendance), and guardian communications.
- Responsible for full-stack integration, connecting the JavaScript frontend to NodeJS backend services, and devised comprehensive documentation for each application.

PROJECTS

OurNigeria ournigeria.ng
Senior Engineer *Feb 2026 - Present*

- Architected the entire platform as an Nx monorepo managing 6 applications: NestJS API backend, Next.js citizen-facing web app (22 chart types with Recharts), Next.js admin dashboard, NestJS document ingestion pipeline, Next.js marketing landing page, and a Remotion video generation service.
- Built a hybrid RAG pipeline combining pgvector similarity search with BM25 full-text search, merged via Reciprocal Rank Fusion (K=60), with Cohere reranking and Voyage AI embeddings (1024-dim, 708K+ chunks across 4 vector indexes), processing 700+ documents across 37 states.
- Implemented SSE streaming for real-time chat responses with keepalive heartbeats every 15s to prevent reverse proxy timeouts, token-budgeted context management (summary + recent messages + user profile), and conversation summarization.
- Integrated Langfuse for LLM observability and scoring, enabling real-time monitoring of AI pipeline performance, token usage, and response quality across all agents.
- Configured CI/CD with GitHub Actions for automated builds and Docker Compose production deployment with multi-stage Dockerfiles, health checks, memory limits (API 1GB, Ingest 4GB, Postgres 2GB), and Playwright E2E tests across 4 test projects.

EDUCATION

University of Nigeria Nsukka, Enugu
BSc Computer Science - 4.1 CGPA *Sep 2019*